PYAE PHYO MAUNG MAUNG

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PROFESSIONAL SUMMARY

Computer Science graduate specializing in software engineering and applied AI systems. Grounded in Object-Oriented Programming (OOP), algorithmic design, and full-stack development, with practical experience architecting LLM-integrated recommendation systems, custom data structures, and machine learning workflows. Proven track record leading technical academic teams and delivering modular web and command-line applications. Eager to contribute scalable software design and robust problem-solving skills to an entry-level software engineering team.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java (JavaSE), PHP, SQL (MySQL), HTML5, CSS3
- **Backend & Frameworks:** FastAPI, Laravel, RESTful APIs, Uvicorn, Pydantic, Bootstrap, Streamlit
- **Generative AI & LLMs:** Groq API, Google Gemini API, Prompt Engineering, Rule-Based Fallback Systems
- **Machine Learning & Data:** Scikit-Learn (TF-IDF, NearestNeighbors), Pandas, NumPy, Neural Networks, Random Forest, Orange3, Data Preprocessing
- **Developer Tools & Platforms:** Git, GitHub, Netlify, XAMPP, StarUML, Figma, WordPress, Microsoft Azure IoT Central
- **Core Concepts & Testing:** Data Structures & Algorithms (BST, Doubly Linked Lists), OOP Design Patterns, UML Modeling, Unittest, Mocking, UI/UX Usability Heuristics
- **Spoken Languages:** English (Fluent), Burmese (Native), Mandarin (Conversational)

TECHNICAL PROJECTS

EcoBite – AI-Powered Sustainable E-Commerce & Recommendation Engine | *Python, FastAPI, Scikit-learn, Pandas, TF-IDF, KNN, Groq API, Google Gemini API, Unittest* **2026**

- Multi-Objective Recommendation Engine: Architected a real-time product recommendation engine using TF-IDF and K-Nearest Neighbors (Cosine Similarity) in Scikit-learn, enabling dynamic multi-objective reranking that balances product semantic similarity with eco-friendliness scores (w_sim vs. w_eco).
- LLM-Driven Nudge System: Integrated Groq (GPT-OSS) and Google Gemini APIs to dynamically generate 1–2 sentence scientifically-grounded 'Eco-Nudges' comparing product lifecycle impacts; implemented a deterministic rule-based fallback mechanism ensuring 100% engine uptime.
- Scalable Data Pipeline: Processed and normalized large-scale datasets from OpenFoodFacts using Pandas chunking (50k batch sizes), handling encoding edge cases, de-duplication, missing image URL reconstruction, and stratified sampling across eco-score grades (A–E).
- High-Performance RESTful API: Built a lightweight FastAPI web service featuring in-memory TF-IDF vector caching at startup, serving sub-millisecond recommendation endpoints and an interactive single-page application (SPA).
- Automated Testing & QA: Developed a unit testing suite using unittest and unittest.mock covering data cleaning functions, vector similarity constraints, API routes, and LLM fallback logic.

Sg Online Library Application (SOLA) | *Python, OOP, Data Structures & Algorithms* **2025**

- Architected a modular CLI library management system in Python 3 enforcing OOP principles (encapsulation, inheritance, polymorphism).
- Implemented a custom Binary Search Tree (BST) keyed by title for logarithmic-time storage, search, removal, and in-order traversal of library catalog items.
- Developed a custom Doubly Linked List to handle dynamic borrower record insertions and removals, integrating linear search algorithms for ID and username lookups.
- Engineered role-based access control (Admin vs. Borrower) mapped via comprehensive UML Class and Use Case diagrams.

Real Estate Commission Management System | *Python, Algorithm Analysis, OOP* **2025**

- Built an automated commission calculation and reporting engine in Python to eliminate manual accounting discrepancies across agent and director tiers.
- Modeled multi-tier inheritance structures linking properties, commercial units, field agents, and executive directors.

- Utilized dynamic array structures to manage sold and unsold inventories, achieving O(1) updates and efficient O(n) iteration for distributed commission statement generation.
- Authored system specifications, encapsulation policies, and validation rules using UML diagrams.

Hotel Management & Booking Platform | *PHP, Laravel, MySQL, Bootstrap* **2024**

- Developed a full-stack hotel reservation platform featuring customer room discovery, booking modules, and an administrative control panel.
- Designed relational database schemas and Entity Relationship Diagrams (ERD) using StarUML; configured local deployments with XAMPP.
- Built dynamic frontend views using Bootstrap, HTML5, CSS3, and JavaScript connected to a Laravel MVC backend.

Diabetes Prediction Engine | *Python, Orange Data Mining, Machine Learning* **2025**

- Preprocessed and normalized the Pima Indians Diabetes dataset, resolving missing parameters and outlier distributions.
- Trained and benchmarked predictive models, achieving 83% accuracy with Neural Networks and 76% accuracy with Random Forest classifiers.

Battery Management System (BMS) IoT Simulation | *Azure IoT Central, System Architecture* **2025**

- Simulated real-time EV battery monitoring for critical telemetry including voltage, current, temperature, State of Charge (SOC), and State of Health (SOH).
- Formulated architectural solutions within Azure IoT Central to mitigate delayed fault detection and fragmented performance logs.

EDUCATION & CERTIFICATIONS

Bachelor of Computing Science | Coventry University (Singapore Campus) **Mar 2025 – Sep 2026**

- Member, Singapore Computer Society (SCS)

Diploma in NCC Level 5 Computing with Business Management | Strategy First International College **Jan 2023 – Jan 2025**

- Academic Standing: GPA 3.90 / 4.00

Diploma in Computing with Business Management | Strategy First International College **Jan 2022 – Dec 2023**

- Academic Standing: GPA 3.90 / 4.00

Certifications

- **AI Foundations – DeepLearning.AI (taught by Andrew Ng)** (Coursera) Mar 2025 – Jul 2025
- **Machine Learning with Python** (Coursera) Jun 2024 – Sep 2024
- **Web Development with PHP & Laravel** (Strategy First International College) Feb 2024 – May 2024

LEADERSHIP & PROFESSIONAL EXPERIENCE

Technical Team Lead (PHP & Laravel Project) | Strategy First International College **Mar 2024 – Jun 2024**

- Directed a software project team through the full lifecycle of a web application from requirements gathering to final deployment.
- Structured milestone sprints, distributed weekly feature backlogs, and coordinated code reviews to facilitate clean Git merges and integration.

Sales Operations & Inventory Lead | Tike War Myanmar **Aug 2020 – Sep 2023**

- Supervised retail and warehouse operations, managing a 3-person team handling genuine machine parts and materials.
- Optimized merchandising and Point of Sale (POS) workflows, contributing to a 30% increase in sales volume.
- Spearheaded weekly social media product marketing initiatives to expand local customer reach.